

Matematik-II Proje Ödevi

Belirli İntegral ile Alan Hesabı - 10 Soru

İçindekiler

- Soru 1: $f(x) = x^2 - 4$, Aralık: $[-2, 3]$
- Soru 2: $f(x) = \sin(x)$, Aralık: $[-\pi, \pi]$
- Soru 3: $y = \sin(x)$ ve $y = \cos(x)$, Aralık: $[0, \pi/2]$
- Soru 4: $y = \sqrt{x}$, $y = 3$, y eksenini
- Soru 5: $y = x^2 - 2$ ve $y = 3x - x^2$
- Soru 6: $y = x^2$, Teğet: $y = 4x - 4$
- Soru 7: $y = x(x^2 - 1) = x^3 - x$
- Soru 8: $f(x) = x \cdot \sin(x)$, Aralık: $[0, \pi]$
- Soru 9: $f(x) = x^4 + 6x^3 + 9x^2 = x^2(x + 3)^2$
- Soru 10: $y = 7e^x$, $y = 3 - 2e^x$, y eksenini

Soru 1: $f(x) = x^2 - 4$, Aralık: $[-2, 3]$

$$f(x) = x^2 - 4$$

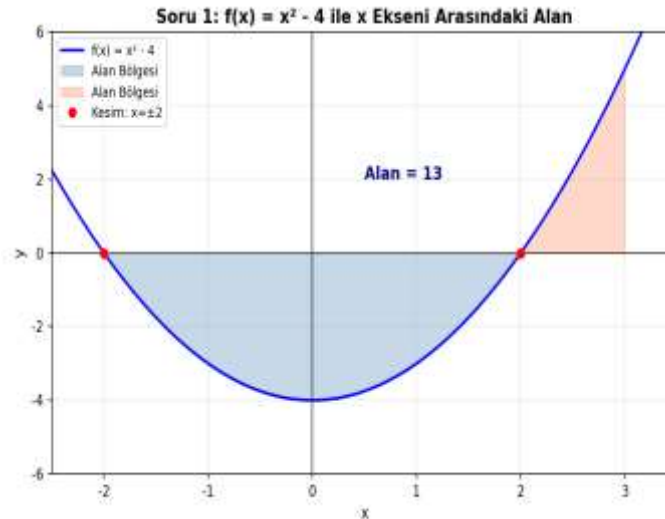
Kesim noktaları: $x^2 - 4 = 0 \rightarrow x = -2, x = 2$

$[-2, 2]$ aralığında $f(x) \leq 0$:

$$\begin{aligned} A_1 &= \left| \int_{-2,2} (x^2 - 4) dx \right| \\ &= \left| \left[\frac{x^3}{3} - 4x \right]_{-2,2} \right| = 32/3 \end{aligned}$$

$[2, 3]$ aralığında $f(x) \geq 0$:

$$A_2 = \int_{2,3} (x^2 - 4) dx = 7/3$$



$$\text{Sonuç: Alan} = A_1 + A_2 = 32/3 + 7/3 = 13$$

Soru 2: $f(x) = \sin(x)$, Aralık: $[-\pi, \pi]$

$f(x) = \sin(x)$, Aralık: $[-\pi, \pi]$

$\sin(x)$ simetrik olduğundan:

$$\text{Alan} = 2 \cdot \int_{[0, \pi]} \sin(x) dx$$

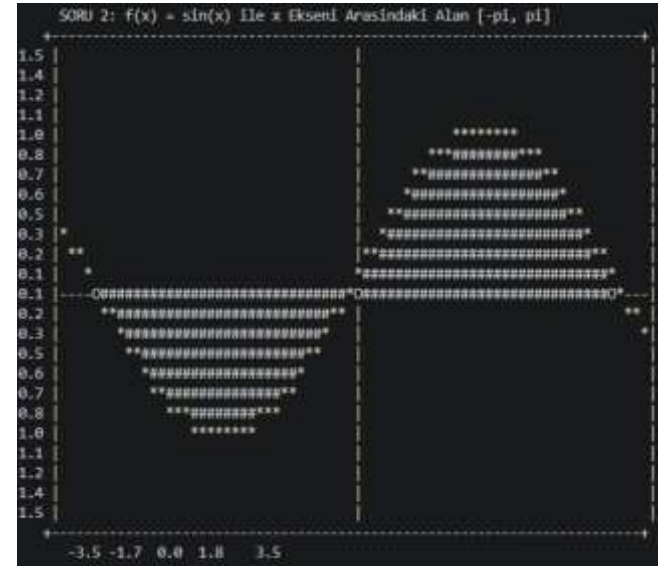
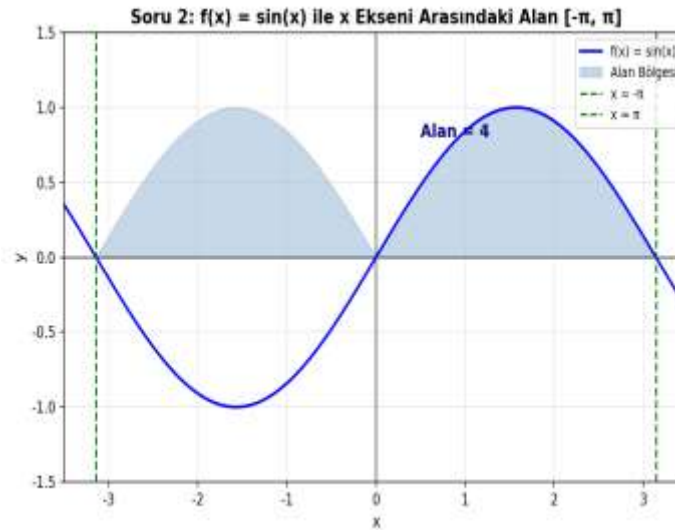
$$\int \sin(x) dx = -\cos(x)$$

$$= 2 \cdot [-\cos(x)]_{[0, \pi]}$$

$$= 2 \cdot (-\cos(\pi) + \cos(0))$$

$$= 2 \cdot (-(-1) + 1)$$

$$= 2 \cdot 2 = 4$$



Sonuç: Alan = 4

Soru 3: $y = \sin(x)$, $y = \cos(x)$, Aralık: $[0, \pi/2]$

$$y_1 = \sin(x), y_2 = \cos(x), [0, \pi/2]$$

Kesim noktası: $\sin(x) = \cos(x)$

$$x = \pi/4$$

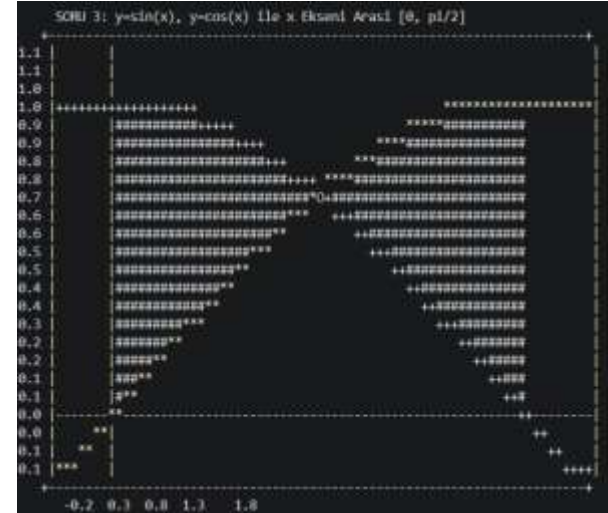
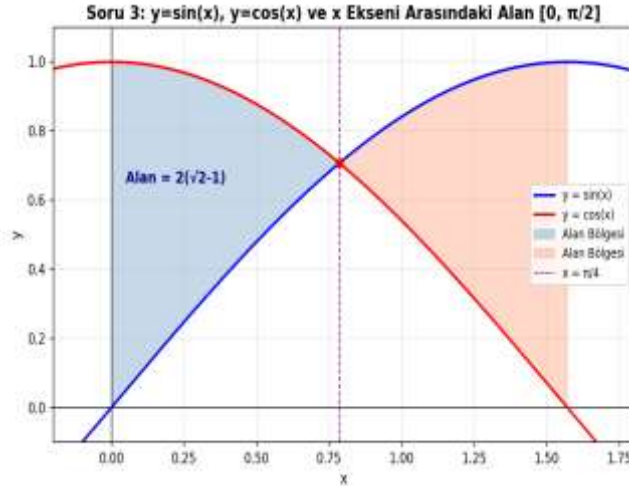
$[0, \pi/4]$ 'te $\cos(x) \geq \sin(x)$:

$$A_1 = \int_{[0, \pi/4]} (\cos - \sin) dx = (\sqrt{2} - 1)$$

$[\pi/4, \pi/2]$ 'de $\sin(x) \geq \cos(x)$:

$$A_2 = \int_{[\pi/4, \pi/2]} (\sin - \cos) dx = (\sqrt{2} - 1)$$

$$\text{Toplam} = 2(\sqrt{2} - 1) \approx 0.8284$$



Sonuç: Alan = $2(\sqrt{2} - 1) \approx 0.8284$

Soru 4: $y = \sqrt{x}$, $y = 3$, y eksen

$y = \sqrt{x}$, $y = 3$, y eksen ($x = 0$)

Kesim: $\sqrt{x} = 3 \rightarrow x = 9$

Sağ üst üçgen alanı:

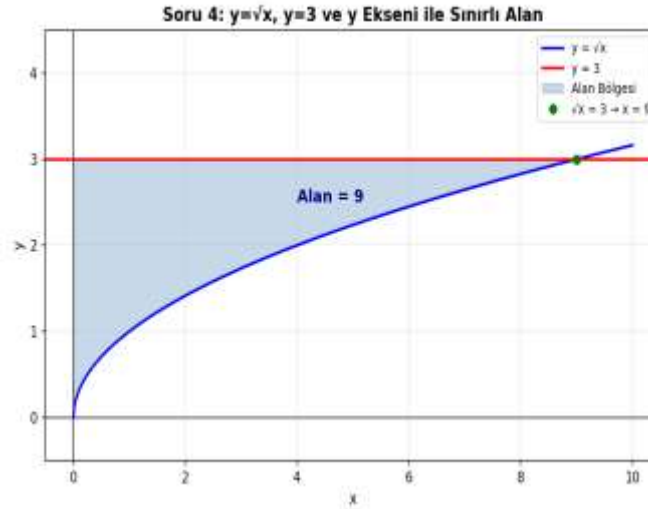
$$\text{Alan} = \int_{[0,9]} (3 - \sqrt{x}) dx$$

$$= [3x - (2/3)x^{3/2}]_{[0,9]}$$

$$= 27 - (2/3)(27)$$

$$= 27 - 18$$

$$= 9$$



Sonuç: Alan = 9

Soru 5: $y = x^2 - 2$ ve $y = 3x - x^2$

$$y_1 = x^2 - 2, \quad y_2 = 3x - x^2$$

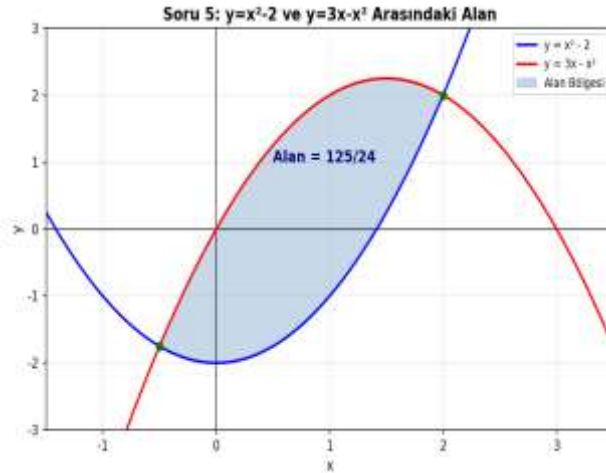
$$\text{Kesim: } x^2 - 2 = 3x - x^2$$

$$2x^2 - 3x - 2 = 0$$

$$(2x+1)(x-2) = 0$$

$$x = -0.5, x = 2$$

$$\begin{aligned} \text{Alan} &= \int_{-0.5}^2 [(3x - x^2) - (x^2 - 2)] dx \\ &= \int_{-0.5}^2 (-2x^2 + 3x + 2) dx \\ &= \left[-\frac{2x^3}{3} + \frac{3x^2}{2} + 2x \right] = \frac{125}{24} \end{aligned}$$



Sonuç: Alan = $\frac{125}{24} \approx 5.2083$

Soru 6: $y = x^2$, Teğet: $y = 4x - 4$

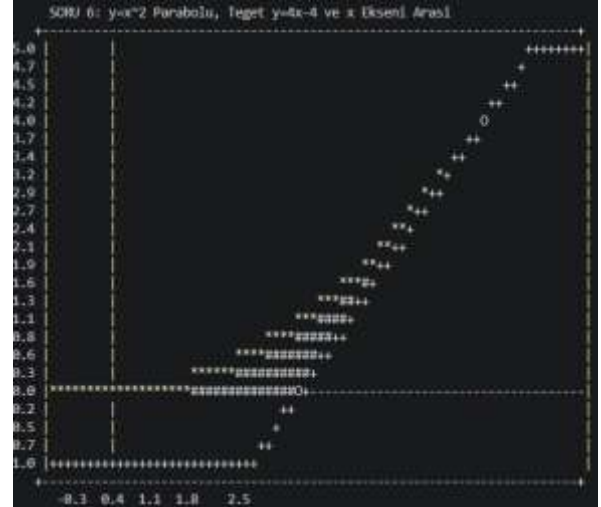
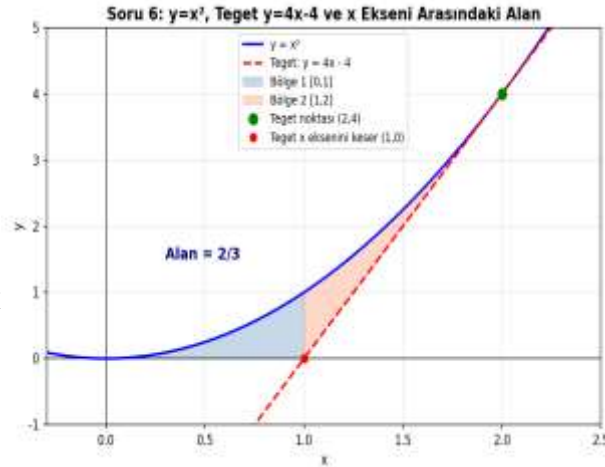
$$y = x^2, \text{ Teğet: } y = 4x - 4$$

$$\text{Teğet noktası: } 2x = 4 \rightarrow x = 2, y = 4$$

$$\text{Teğet x eksenini keser: } 4x - 4 = 0 \rightarrow x = 1$$

$$\text{Parabol x eksenini keser: } x^2 = 0 \rightarrow x = 0$$

$$\begin{aligned} \text{Alan} &= \int_{[0,1]} x^2 dx + \int_{[1,2]} (x^2 - (4x - 4)) dx \\ &= [x^3/3]_{[0,1]} + [-5x + x^3/3 + 4x]_{[1,2]} \\ &= 1/3 + 1/3 = 2/3 \end{aligned}$$



Sonuç: Alan = 2/3

Soru 7: $y = x(x^2-1) = x^3-x$

$$y = x^3 - x = x(x-1)(x+1)$$

Kökler: $x = -1, x = 0, x = 1$

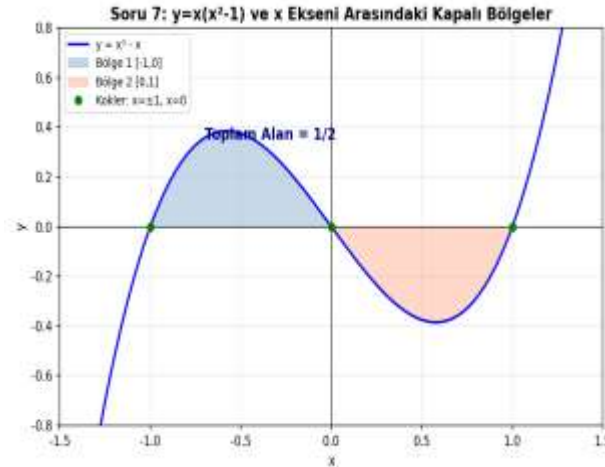
$[-1, 0]$ aralığında $f(x) \geq 0$:

$$A_1 = \int_{-1,0} (x^3-x) dx = 1/4$$

$[0, 1]$ aralığında $f(x) \leq 0$:

$$A_2 = |\int_{0,1} (x^3-x) dx| = 1/4$$

$$\text{Toplam} = A_1 + A_2 = 1/2$$



Sonuç: Alan = 1/2

Soru 8: $f(x) = x \sin(x)$, Aralık: $[0, \pi]$

$$f(x) = x \cdot \sin(x), [0, \pi]$$

Parçalı integral (IBP):

$$u = x \rightarrow du = dx$$

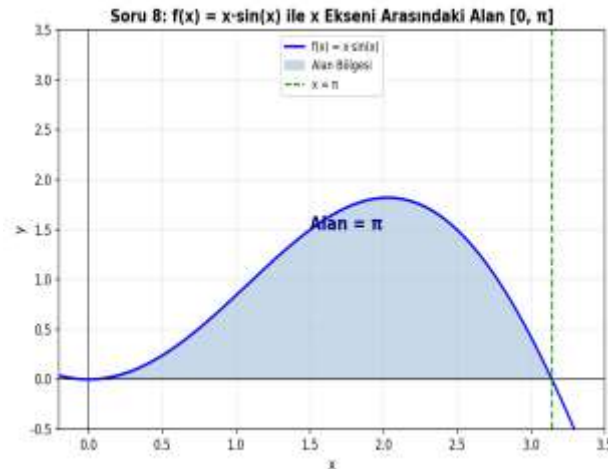
$$dv = \sin(x)dx \rightarrow v = -\cos(x)$$

$$\int_{[0, \pi]} x \cdot \sin(x) dx$$

$$= [-x \cdot \cos(x)]_{[0, \pi]} + \int_{[0, \pi]} \cos(x) dx$$

$$= (-\pi \cdot (-1) - 0) + [\sin(x)]_{[0, \pi]}$$

$$= \pi + 0 = \pi$$



Sonuç: Alan = $\pi \approx 3.1416$

Soru 9: $f(x) = x^4 + 6x^3 + 9x^2 = x^2(x+3)^2$

$$f(x) = x^4 + 6x^3 + 9x^2 = x^2(x+3)^2$$

Kökler: $x = -3$ (çift), $x = 0$ (çift)

→ Eğri x eksenine teğet

→ $f(x) \geq 0$ tüm x için

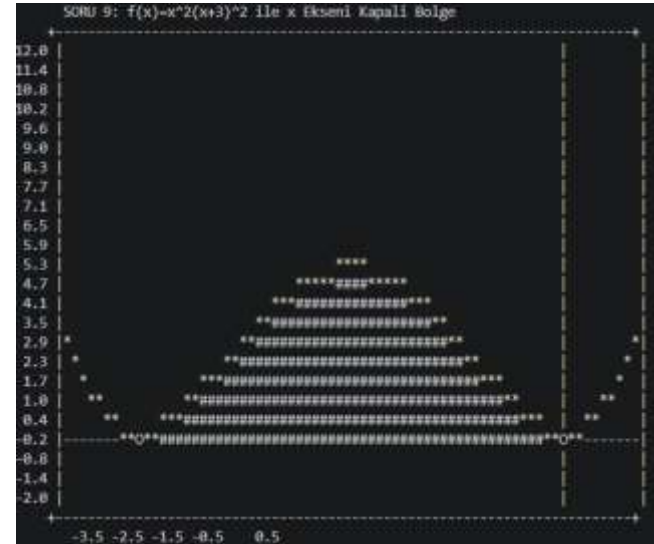
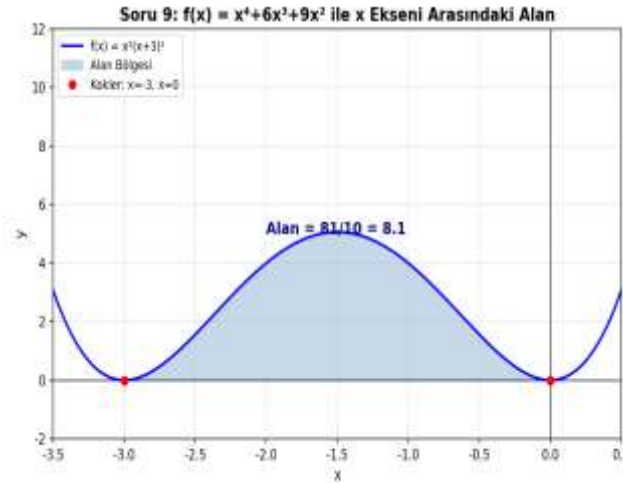
$$\text{Alan} = \int_{-3,0} x^2(x+3)^2 dx$$

$$= \int_{-3,0} (x^4 + 6x^3 + 9x^2) dx$$

$$= [x^5/5 + 6x^4/4 + 9x^3/3]_{-3,0}$$

$$= |(0) - (-243/5 + 243/2 - 27)|$$

$$= 81/10 = 8.1$$



Sonuç: Alan = 81/10 = 8.1

Soru 10: $y = 7e^x$, $y = 3 - 2e^x$, y eksen

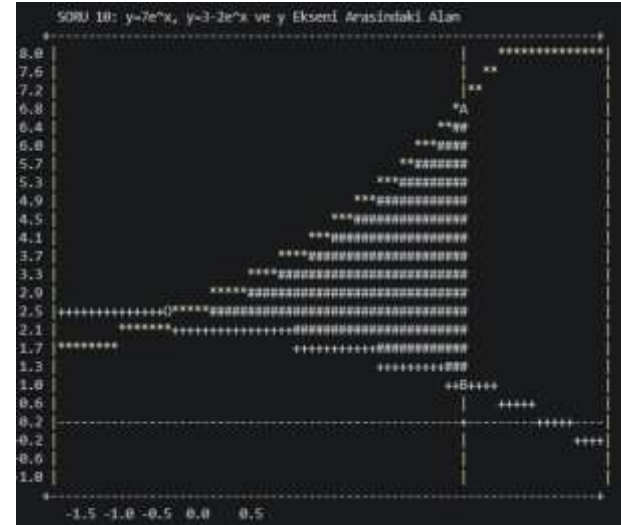
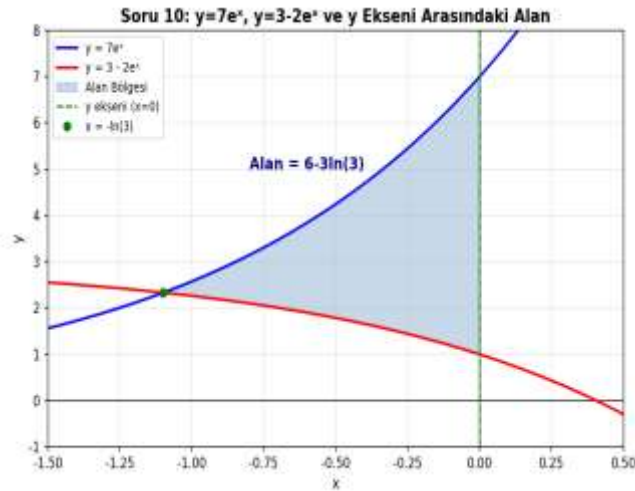
$y_1 = 7e^x$, $y_2 = 3 - 2e^x$, y eksen ($x = 0$)

Kesim: $7e^x = 3 - 2e^x$

$$9e^x = 3 \rightarrow e^x = 1/3$$

$$x = -\ln(3)$$

$$\begin{aligned}\text{Alan} &= \int_{[-\ln 3, 0]} [(3 - 2e^x) - 7e^x] dx \\ &= \int_{[-\ln 3, 0]} (3 - 9e^x) dx \\ &= [3x - 9e^x] = (0 - 9) - (-3\ln 3 - 3) \\ &= -9 + 3\ln 3 + 3 = 3\ln 3 - 6 \\ &= 6 - 3\ln(3) \approx 2.7042\end{aligned}$$



Sonuç: Alan = $6 - 3\ln(3) \approx 2.7042$

Teşekkürler

